

Volume IV · Chapter 22 — State of the Art & Evaluation (text/image/video) · Theory

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-IV-Chapter-22-context.md` for the AI interaction log.

Prerequisites: Ch 10, Ch 15, Ch 19.

Learning outcomes — the student can: survey current medical LLM/multimodal/video SOTA honestly; design evaluation for book-derived models; assess source-fidelity and citation correctness; articulate maturity gaps.

Topics (theory) — 8 h:

#	Topic	h
22.1	Text medical LLMs today: Med-PaLM 2, Meditron, MedGemma; capabilities	2
22.2	Multimodal (image+text): Med-PaLM M, LLaVA-Med; the reliability gap	2
22.3	Video understanding: surgical/echo/endoscopy — early state	2
22.4	Evaluating book-derived models: source-fidelity & citation correctness	2

Datasets/tools: eval harness; the RAG/fine-tuned models built in Volume IV. **Assessment:** SOTA briefing + fidelity eval (**60%**); citation-accuracy report (**20%**); quiz (**20%**). **Key decisions:** honest capability claims; which modality is production-ready; how to report limits. **References:** plan §6.3; §13 → *Medical LLMs*. **Hours:** Theory **8** + Lab **6** = **14**.